ClipSAT : Math for AI & Data Science

Unit 8.01: PCA & Dimensionality Reduction

Name: _____

Q1. Principal Component Analysis PCA projects a dataset from mathbb{R}^{30} onto its top 2 principal components. What is the dimension of each transformed data point?

[ ] A: 15
[ ] B: 30
[ ] C: 28
[ ] D: 2

Q2. Principal Component Analysis PCA projects a dataset from mathbb{R}^{40} onto its top 4 principal components. What is the dimension of each transformed data point?

[ ] A: 20
[ ] B: 40
[ ] C: 36
[ ] D: 4

Q3. Principal Component Analysis PCA projects a dataset from mathbb{R}^{50} onto its top 6 principal components. What is the dimension of each transformed data point?

[ ] A: 25
[ ] B: 50
[ ] C: 44
[ ] D: 6

Q4. Principal Component Analysis PCA projects a dataset from mathbb{R}^{60} onto its top 8 principal components. What is the dimension of each transformed data point?

[ ] A: 30
[ ] B: 60
[ ] C: 52
[ ] D: 8

Q5. Principal Component Analysis PCA projects a dataset from mathbb{R}^{70} onto its top 10 principal components. What is the dimension of each transformed data point?

[ ] A: 10
[ ] B: 70
[ ] C: 60
[ ] D: 35

Q6. Principal Component Analysis PCA projects a dataset from mathbb{R}^{80} onto its top 12 principal components. What is the dimension of each transformed data point?

[ ] A: 12
[ ] B: 80
[ ] C: 68
[ ] D: 40

Q7. Principal Component Analysis PCA projects a dataset from mathbb{R}^{90} onto its top 14 principal components. What is the dimension of each transformed data point?

[ ] A: 14
[ ] B: 90
[ ] C: 76
[ ] D: 45

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Q8. Principal Component Analysis PCA projects a dataset from mathbb{R}^{100} onto its top 16 principal components. What is the dimension of each transformed data point?

[ ] A: 100

[ ] B: 16

[ ] C: 84

[ ] D: 50